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October 30, 2026

Predicting canine height beyond admixture

Introduction

Body size is among the most variable traits in domestic dogs, spanning from ankle-high Chihuahuas to Great Danes that can stand nearly hip-high to their owners.

Much of that variation traces to a small number of loci of large effect, including IGF1, LCORL, GHR, HMGA2, and SMAD2, which have been identified largely through comparisons among purebred dogs. Because breed identity is so strongly tied to size, however, genetic associations with height in a mixed population are difficult to separate from breed ancestry. A genetic marker can predict height simply because it indicates which breeds a dog descends from, without carrying any information beyond ancestry.

Morrill et al. (2022) addressed this using a cohort of dogs from Darwin's Ark. They identified size-associated SNPs ($p < 1e\text{-}5$) and trained a random forest on 2,733 of them in 1,730 dogs, which predicted size well in held-out dogs (Pearson $r \approx 0.77$, with about 66% of predictions within ±0.5 units) and outperformed random SNP sets. (They define size on an ordinal scale 1 through 5 based on the size question answers, i.e., is your dog: Ankle high or shorter, Calf high, Knee high, Thigh high, Hip high or higher. This convention is used in this analysis). Applying the same approach to a larger Darwin's Ark cohort, I obtained similar performance (random forest on 2,508 SNPs with $p < 1e\text{-}5$, trained on 2,025 dogs and tested on 507 held-out dogs: $r = 0.77$, 67% within ±0.5 units). Their results show that genotype data predict size in mixed-breed dogs, but they leave open how much of that prediction is attributable to breed ancestry.

This report asks whether genome-wide genetic information predicts height beyond what breed admixture already tells us. Using a cohort of 3,277 dogs from Darwin's Ark (2,532 retained after excluding dogs with missing covariates or height and dogs under 1 year old; 2,025 train, 507 test), I compare predictors built from breed admixture and covariates (sex, neutered status, age) with a genome-wide polygenic score (PGS) estimated by BayesR on training dogs only. I then test whether the score's association with height persists after both are adjusted for ancestry, and ask how much of that association comes from established size loci versus the rest of the genome.

Models predicting height with admixture, covariates, and PGS predictors

I first compared how well each predictor explained height in the held-out test dogs ($n \approx 500$). Breed admixture alone explained 52.4% of the variance in height ($R^2 = 0.524$), and adding sex, neutered status, and age raised this to 0.549. A genome-wide polygenic score (PGS) alone achieved $R^2 = 0.588$, and the PGS with covariates reached 0.601. The PGS with covariates

outperformed admixture with covariates by $\Delta R^2 = +0.051$ (95% bootstrap CI 0.009 to 0.096), a modest but significant increase. Adding admixture to the PGS and covariates raised R^2 only to 0.604 ($\Delta R^2 = +0.003$), suggesting that, once the PGS is included, ancestry contributes little additional predictive information.

The coefficients for the models were not fit on the training dogs. There the PGS is overfit and looks far more predictive than it is on new dogs, so a model fit on training dogs learns coefficients for the other terms that compensate for a PGS that won't be that strong later. In a preliminary analysis that fit all coefficients on training dogs, adding admixture to the PGS and covariates lowered test R^2 (0.593 to 0.577), which I hypothesized was the result of PGS being overfit. Instead, the combining weights were fit within the test dogs by 10-fold cross-validation. For each fold, weights were estimated on nine-tenths of the test dogs and used to predict the remaining tenth, so every test dog's prediction comes from weights learned without that dog. R^2 and the other metrics were computed from these out-of-fold predictions. Each model has at most five fitted weights, which keeps the risk of overfitting at ~500 dogs low.

Table 1: Test-set R^2 for height prediction models using genetic ancestry, a polygenic score (PGS), and covariates.

Model	R^2
Admixture	0.524
Admixture+covariates	0.549
PGS	0.588
PGS+covariates	0.601
PGS+covariates+admixture	0.604

A genome-wide score predicts height at least as well as breed admixture, and modestly better ($\Delta R^2 = +0.05$, 95% CI 0.01 to 0.10).

Is PGS just capturing breed ancestry?

Because the PGS was estimated without adjusting for ancestry, it is expected to carry breed information. To quantify this, I regressed the PGS on breed admixture and covariates (sex, neutered status, age) in the training dogs and evaluated the fit in the test dogs. These variables explained 83% of the variance in the PGS ($R^2 = 0.829$ in test dogs), making PGS a largely breed-driven quantity.

But a high ancestry share does not mean the remaining variation is uninformative, and it does not mean the PGS adds nothing beyond ancestry. I explored whether the part of the PGS not explained by ancestry still tracks the part of height not explained by ancestry. Height and the PGS were each regressed on admixture and covariates in the training dogs, and the residuals

were computed in the test dogs. These residuals were correlated at r = 0.375 (95% bootstrap CI 0.267 to 0.476; $r^2 = 0.14$). The component of the PGS that ancestry cannot account for thus still explains about 14% of the variation in height that ancestry cannot account for.

These results show that although the PGS mostly reflects breed ancestry it is not only a breed proxy. The next section asks where in the genome this residual association is coming from.

Is predictive signal coming from known and established size loci, or elsewhere?

Body size in dogs is often described as being controlled by a small number of loci of large effect, including IGF1, LCORL, HMGA2, GHR, and SMAD2. In a mixed-breed cohort, a marker near one of these genes could therefore predict height simply because it indicates breed ancestry. At the same time, a genome-wide score might capture additional contributions from the rest of the genome. I sought to discover how much of the PGS's association with height, after adjustment for ancestry, comes from previously reported size loci and how much from the remainder of the genome.

To answer this question, I split the PGS into two scores using the same BayesR SNP effects. The known-loci score sums the effects of SNPs within ±500 kb of previously reported size loci (IGF1, GHR, LCORL, HMGA2, IGF2BP2, IGF1R, SMAD2, MSRB3, MED13L, ANAPC1, SAR1B), and the rest score sums all other SNPs. 3,127 out of 408,230 SNPs were located within the known-loci windows. The results (shown in Table 2 below) indicate that both the known-loci score and the rest score remain associated with height after accounting for admixture and covariates.

Table 2: Residual correlation with height by PGS component. All p-values < 8.2e-06

PGS Component	Residual correlation with height
Full PGS	0.375
Established-locus PGS	0.353
Remaining-genome PGS	0.197

How Is Genetic Signal Distributed Across the Rest of the Genome?

To investigate how height-associated genetic signal was distributed outside the established size-associated regions, SNPs in the remainder of the genome were ranked by their posterior inclusion probability (PIP) from BayesR. PIP measures the posterior probability that a SNP belongs to a component with a nonzero effect, providing an indication of the strength of evidence that the variant contributes to the model. Ranking SNPs by PIP identifies variants with the strongest evidence of association within the rest-of-genome component.

The remaining SNPs were divided into three groups based on their BayesR posterior inclusion probabilities (PIPs). The top_1000 group contained the 1,000 highest-PIP SNPs, the next_9000

group contained the next 9,000 highest-PIP SNPs, and the bulk group contained all remaining SNPs outside the established regions. I examined the association between each group's PGS and height after accounting for genetic ancestry and covariates. All three groups retained a positive, statistically significant association with height, shown in Table 3 below.

Table 3: Residual correlation with admixture-adjusted height for PGS built from known-locus windows versus other SNP bins. All p-values < 0.001

Bin	SNPs	Residual r	r^2
Known-locus windows (±500 kb)	3,152	0.353	0.125
Top 1k	1,000	0.160	0.026
Next 9k	9,000	0.155	0.024
Bulk	395,078	0.150	0.023

The similar correlation estimates suggest that height-associated signal beyond modeled ancestry and covariates is not confined to the highest-PIP SNPs. The next 9,000 SNPs and the much larger bulk group also retain associations with height. Although the top 1,000 SNPs had the largest estimated correlation, the differences between the three estimates were small, and these results do not establish that one group has a significantly stronger association than the others. Overall, the findings are consistent with height-associated signal being distributed across many SNPs outside the established size-associated regions, rather than being concentrated exclusively in the highest-PIP variants.

How much of the explained residual variation comes from known loci versus the rest of the genome?

To quantify the relative contributions of established size-associated genomic regions and the remainder of the genome, I used Shapley-value decomposition of the variance explained by their respective PGS components. Both components were residualized against genetic ancestry and covariates before analysis, allowing me to examine their associations with height beyond these factors. The Shapley method allocates the joint model's explained variance between the two components by averaging each component's incremental contribution across the two possible orders in which they can be added to the model.

Table 4

Component	r^2	95% CI	Share of joint r^2
Joint (known + rest)	0.150		100%
Known loci (Shapley)	0.118	[0.057, 0.192]	79% [59,92]
Rest of genome (Shapley)	0.032	[0.010, 0.065]	21% [8,41]

These results suggest that established size-associated regions account for the larger share of the height-associated signal captured by the two PGS components in this analysis. However, the rest-of-genome component also contributed to the joint model, indicating that height-associated genetic signal is not restricted to the established regions.

Conclusion

I asked whether genome-wide genetic information predicts height beyond what breed admixture already tells us. In held-out dogs, a polygenic score (PGS) with covariates explained modestly more height variance than admixture with covariates ($\Delta R^2 = +0.051$, 95% CI 0.009 to 0.096), and adding admixture to the PGS and covariates added little (R^2 0.601 to 0.604). The PGS is therefore at least as good a predictor as ancestry, and somewhat better.

The PGS is largely a breed-driven quantity: admixture and covariates explained 83% of its variance. It is not only a proxy for ancestry, however. After both height and the PGS were adjusted for admixture and covariates, they remained correlated ($r = 0.375$), so the component of the score that ancestry does not explain still tracks the component of height that ancestry does not explain.

Most of this residual association came from established size loci. The known-loci score was correlated with adjusted height ($r = 0.35$), and about four-fifths of the explained residual variation was attributable to these loci (79%, 95% CI 59 to 92%). A smaller association remained outside them ($r \approx 0.20$ under linear adjustment), and it was not concentrated in the highest-confidence SNPs, so no small set of additional loci accounts for it.

These results support a picture in which known size loci carry most of the height information that is independent of breed, and the rest of the genome contributes a smaller, diffuse component.